

TATA STEEL: Digital Guide Knowledge Base

Technical SOPs & Safety Guidelines | Internal Training Document

1. Blast Furnace-4 (BF-4) Operations

The Blast Furnace is the heart of iron making. Precision in temperature and pressure management is critical.

Standard Operating Temperature: 1200°C to 1500°C.

Critical Safety Limit: 1550°C. If exceeded, immediate alert to Control Room is mandatory.

Technical Specs (For Engineers): BF-4 utilizes a Top Gas Recovery Turbine (TRT). Critical pressure threshold is 2.5 bar. GCP efficiency must be maintained at 99%.

2. Safety Protocols (LOTO & PPE)

SAFETY FIRST: Zero Tolerance Policy

- **PPE Mandatory:** Helmet, Steel-toed boots, High-visibility jacket, Safety goggles.
- **Lockout-Tagout (LOTO):** Before any maintenance on Conveyor Belt 3, the power isolation switch must be locked with a personal tag.
- **Emergency Assembly:** In case of gas leak, proceed to Point 4 Assembly Area immediately.

3. Maintenance: Conveyor Belt 3

Daily Inspection: Check for belt misalignment and abnormal motor noise.
Lubrication: Apply Grease-Grade 2 to all rollers every 24 hours of operation.

4. Troubleshooting Guide

Problem	Immediate Action (Junior)	Technical Root Cause (Senior)
Low Oxygen Flow	Notify supervisor; check valve 'V-10'.	Oxygen Lance blockage; check pressure set at 10kg/cm2.